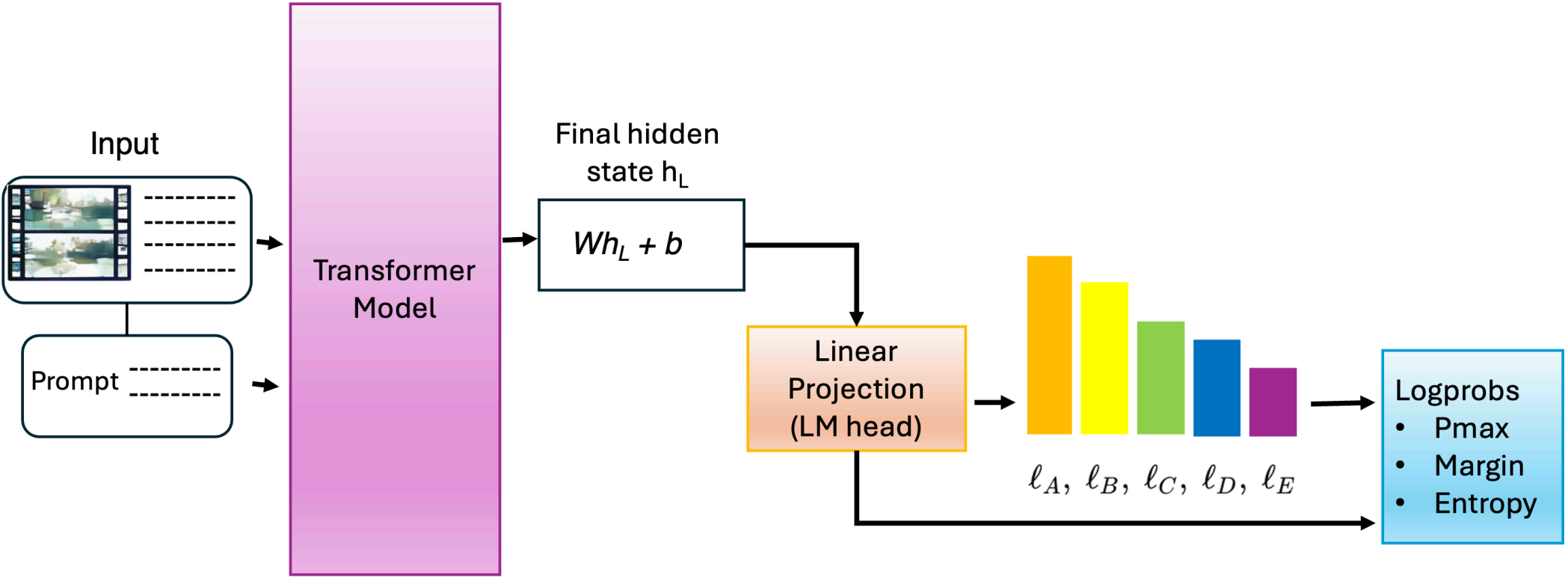

$$\log p_i = \ell_i - \log \sum_{k \in \{A,B,C,D,E\}} e^{\ell_k}$$